

PWA Force Update System — Backend Documentation

Overview

The frontend PWA automatically sends the current application version and platform information in every API request using request headers. Backend validates the version and can force the client to update when required. This system is mainly used for breaking backend/API changes, security patches, invalid cached frontend builds, critical production bugs, database/schema migrations, and mandatory feature rollouts.

Request Headers

Frontend automatically sends:

x-app-version: 1.0.3

x-platform: pwa

Backend Validation Flow

Request received → Read x-app-version → Compare against minimum supported version → If outdated return HTTP 426 Upgrade Required → Else continue request normally.

Recommended Backend Response

HTTP Status: **426 Upgrade Required**

JSON Response:

```
{ "code": "APP_UPDATE_REQUIRED", "message": "Please update Elle Hub to continue." }
```

Frontend Update Behavior

When frontend receives HTTP 426 or APP_UPDATE_REQUIRED, the frontend automatically checks for the latest PWA build, shows update UI, downloads latest assets, activates latest service worker, and reloads the application safely.

Version Format

Recommended version format: MAJOR.MINOR.PATCH

Examples: 1.0.3, 1.1.0, 2.0.0

Suggested Backend Configuration

```
{ "minimum_supported_version": "1.0.3", "latest_version": "1.1.0" }
```

Recommended Force Update Use Cases

Breaking backend/API changes, security patches, invalid cached frontend builds, critical frontend production bugs, and database/schema migrations.

Do NOT Force Update For

Minor UI changes, text fixes, analytics updates, optional features, and non-breaking frontend improvements.

Suggested Backend Middleware Logic

Read x-app-version header → Compare with minimum_supported_version → If outdated return 426 APP_UPDATE_REQUIRED → Else continue request.

Platform Header

Frontend also sends x-platform: pwa. Possible future values include pwa, ios, android, and desktop. This allows platform-specific update validation in the future.

Maintenance Mode

Use HTTP 503 Service Unavailable with response: { "code": "MAINTENANCE_MODE", "message": "System temporarily unavailable." }

Important Notes

Backend should NOT trigger frontend reloads directly, inject scripts, force browser redirects, or invalidate sessions unnecessarily. Backend only needs to return proper HTTP status codes and JSON responses. The frontend PWA system already handles service worker updates, cached asset replacement, stale cache cleanup, update UI, safe

activation, and application reload flow.

Recommended HTTP Status Codes

Status Code	Purpose
426	Force application update
503	Maintenance mode
401	Authentication expired
403	Permission denied
409	Version conflict

Final Update Flow

Frontend sends app version → Backend validates version → Backend returns 426 if outdated → Frontend detects update requirement → Update UI appears → Latest PWA build installs → Application reloads safely.